Sieve of Eratosthenes (1–100)

Mathematical description. Fix $n = 100$. Let $S = \{2, 3, \ldots, n\}$. For each integer p defined as the smallest element of S not yet processed, declare p *prime* and remove from S every integer of the form kp with $k \geq 2$. Continue this procedure while $p^2 \leq n$. The elements of S that remain after this process are precisely the prime numbers $\leq n$.

Algorithm (short).

1. Initialize the list $2, 3, \ldots, n$.
2. For each current smallest unprocessed p (starting at $p = 2$), remove all multiples $2p, 3p, 4p, \ldots$ up to n.
3. Stop once $p^2 > n$ (i.e. $p > \lfloor \sqrt{n} \rfloor$). Remaining numbers are primes.

Note. 1 is neither prime nor composite.

	2	**3**	4	**5**	6	**7**	8	9	10
11	12	**13**	14	15	16	**17**	18	**19**	20
21	22	**23**	24	25	26	27	28	**29**	30
31	32	33	34	35	36	**37**	38	39	40
41	42	**43**	44	45	46	**47**	48	49	50
51	52	**53**	54	55	56	57	58	**59**	60
61	62	63	64	65	66	**67**	68	69	70
71	72	**73**	74	75	76	77	78	**79**	80
81	82	**83**	84	85	86	87	88	**89**	90
91	92	93	94	95	96	**97**	98	99	100

Primes The primes ≤ 100 are:

2, **3**, **5**, **7**, **11**, **13**, **17**, **19**, **23**, **29**, **31**, **37**, **41**, **43**, **47**, **53**, **59**, **61**, **67**, **71**, **73**, **79**, **83**, **89**, **97**.

0.1 L1: 1.1 Divisibility (syllabus, expectations)

0.2 L2: 1.2 Prime numbers (open conjectures, breakthroughs)

0.3 L3: 1.3 GCD, 1.4 Euclidean Algorithm

0.4 L4: 1.5 Fundamental Theorem of Arithmetic

0.5 L5: 1.5 Dirichlet's Theorem, 2.1 Congruences

Lecture 5: Dirichlet's Theorem and applications of the FTA

§ Theorem 1.21 (Dirichlet's theorem on primes in arithmetic progressions) Let $a, b \in \mathbb{Z}$ with $b > 0$ and $\gcd(a, b) = 1$. Then the arithmetic progression

$$a,\ a+b,\ a+2b,\ a+3b,\ \dots$$

contains infinitely many prime numbers.

§ Remark (on Theorem Theorem 1.21 (Dirichlet's theorem on primes in arithmetic progressions)) Setting $a = 1$ and $b = 1$ recovers the elementary statement that there are infinitely many primes. (We remark that the full proof of Dirichlet's theorem requires analytic tools and is beyond the scope of this course.)

§ Proposition 1.22 (Infinitely many primes congruent to 3 modulo 4) There are infinitely many primes of the form $4n + 3$, where n is a nonnegative integer.

§ Lemma 1.23 (Product of numbers congruent to 1 modulo 4) Let $a, b \in \mathbb{Z}$. If $a \equiv 1 \pmod 4$ and $b \equiv 1 \pmod 4$, then $ab \equiv 1 \pmod 4$.

Proof. Write $a = 4n_1 + 1$ and $b = 4n_2 + 1$ with $n_1, n_2 \in \mathbb{Z}$. Then

$$ab = (4n_1 + 1)(4n_2 + 1) = 16n_1n_2 + 4n_1 + 4n_2 + 1 = 4(4n_1n_2 + n_1 + n_2) + 1,$$

so $ab \equiv 1 \pmod 4$. This proves the lemma. □

Proof of Proposition Proposition 1.22 (Infinitely many primes congruent to 3 modulo 4). Suppose, for contradiction, that there are only finitely many primes congruent to 3 modulo 4. List them as

$$P_1 = 3,\ P_2,\ P_3,\ \dots,\ P_r,$$

all distinct and each $P_i \equiv 3 \pmod 4$.

Consider the integer

$$N := 4P_1P_2 \cdots P_r - 1.$$

Note that $N \equiv -1 \equiv 3 \pmod 4$.

Let q be any prime divisor of N. We make two observations:

1. For each $i = 1, \dots, r$ we have $N \equiv -1 \pmod{P_i}$, so $P_i \nmid N$. Thus q is not equal to any of the P_i.
2. Since $N \equiv 3 \pmod 4$, it is not possible that every prime divisor of N is congruent to 1 modulo 4. Indeed, a product of integers each congruent to $1 \pmod 4$ is itself congruent to $1 \pmod 4$ by Lemma Lemma 1.23 (Product of numbers congruent to 1 modulo 4), contradicting $N \equiv 3 \pmod 4$. Hence some prime divisor q of N satisfies $q \equiv 3 \pmod 4$.

Combining the two observations, q is a prime congruent to 3 modulo 4 that is not among $P_1, \dots, P_r$, contradicting the assumed finiteness of such primes. Therefore there are infinitely many primes congruent to 3 modulo 4. □

§ Remark (on Proposition Proposition 1.22 (Infinitely many primes congruent to 3 modulo 4)) The method above—building an integer with residue $3 \pmod 4$ whose prime divisors cannot all be from a given finite list—gives a proof in this special case of Dirichlet's theorem. However, this argument does not extend directly to all arithmetic progressions; for the full theorem one needs deeper tools (Dirichlet characters and L-series).

Congruences

§ Historical remark (Gauss and congruences) Up to the beginning of the nineteenth century number theory consisted of many isolated results. In 1801, Gauss, in his *Disquisitiones Arithmeticae*, introduced the modern theory of congruences and unified many isolated results in number theory.

§ Definition (Congruence modulo m) Let $a, b, m \in \mathbb{Z}$ with $m > 0$. We say that *a is congruent to b modulo m*, and write

$$a \equiv b \pmod{m},$$

if $m \mid (a - b)$. The integer m is called the *modulus* of the congruence.

§ Remark (notation) The notation $a \not\equiv b \pmod{m}$ means $m \nmid (a - b)$. The modulus is customarily taken to be positive; we will assume $m > 0$ in the sequel.

§ Proposition 2.1 (Congruence modulo m is an equivalence relation) Congruence modulo m is reflexive, symmetric, and transitive on $\mathbb{Z}$; hence it is an equivalence relation.

Proof. Reflexive: For any $a \in \mathbb{Z}$, $a - a = 0$ and $m \mid 0$, so $a \equiv a \pmod{m}$.

Symmetric: If $a \equiv b \pmod{m}$ then $m \mid (a - b)$, hence $m \mid -(a - b) = (b - a)$, so $b \equiv a \pmod{m}$.

Transitive: If $a \equiv b \pmod{m}$ and $b \equiv c \pmod{m}$ then $m \mid (a - b)$ and $m \mid (b - c)$, therefore $m \mid (a - b) + (b - c) = a - c$, so $a \equiv c \pmod{m}$. □

§ Consequence 2.2 (Partition into residue classes) The set $\mathbb{Z}$ is partitioned into equivalence classes under congruence modulo m.

§ Definition (Notation for residue class) For $x \in \mathbb{Z}$, write $[x]_m$ (or simply $[x]$ when m is understood) for the equivalence class of x modulo m:

$$[x]_m = \{y \in \mathbb{Z} : y \equiv x \pmod{m}\}.$$

(Do not confuse $[x]_m$ with the floor function.)

Example: The equivalence classes modulo 4

$$\begin{aligned}
[0]_4 &= \{x \in \mathbb{Z} : x \equiv 0 \pmod{4}\} = \{4n : n \in \mathbb{Z}\} = \{\dots, -8, -4, 0, 4, 8, \dots\},\\
[1]_4 &= \{x : x = 4n + 1,\ n \in \mathbb{Z}\} = \{\dots, -7, -3, 1, 5, 9, \dots\},\\
[2]_4 &= \{x : x = 4n + 2,\ n \in \mathbb{Z}\} = \{\dots, -6, -2, 2, 6, 10, \dots\},\\
[3]_4 &= \{x : x = 4n + 3,\ n \in \mathbb{Z}\} = \{\dots, -5, -1, 3, 7, 11, \dots\}.
\end{aligned}$$

Thus $\mathbb{Z}$ is partitioned into four residue classes modulo 4, and every integer is congruent to exactly one of $0, 1, 2, 3$ modulo 4.

§ Definition (Complete residue system modulo m) A set $R \subset \mathbb{Z}$ is a *complete residue system modulo m* if every integer is congruent modulo m to exactly one element of R.

§ Example (complete residue systems) The set $\{0, 1, 2, \dots, m - 1\}$ is a complete residue system modulo m. Also, for example, $\{6, -11, 19, 1988\}$ is a complete residue system modulo 4, since $6 \equiv 2$, $-11 \equiv 1$, $19 \equiv 3$, and $1988 \equiv 0 \pmod{4}$.

§ Proposition 2.3 (Standard complete residue system) The set $\{0, 1, 2, \dots, m - 1\}$ is a complete residue system modulo m.

Proof. Let $a \in \mathbb{Z}$. By the Division Algorithm there exist unique $q, r \in \mathbb{Z}$ with $0 \leq r < m$ such that $a = mq + r$. Then $a - r = mq$, so $m \mid (a - r)$ and $a \equiv r \pmod{m}$, with $r \in \{0, 1, \ldots, m-1\}$. This shows every integer is congruent to at least one element of $\{0, 1, \ldots, m-1\}$.

To show uniqueness: suppose $r_1, r_2 \in \{0, 1, \ldots, m-1\}$ and $r_1 \equiv r_2 \pmod{m}$. Then $m \mid (r_1 - r_2)$, but $-(m-1) \leq r_1 - r_2 \leq m - 1$. The only multiple of m in this range is 0, so $r_1 - r_2 = 0$ and hence $r_1 = r_2$. Thus each integer is congruent to exactly one element of $\{0, 1, \ldots, m-1\}$. □

§ Proposition 2.4 (Arithmetic with congruences) Let $a, b, c, d \in \mathbb{Z}$ and $m > 0$. If $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$, then

1. $a + c \equiv b + d \pmod{m}$,
2. $ac \equiv bd \pmod{m}$.

Proof. From $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$ we have $m \mid (a - b)$ and $m \mid (c - d)$.

For (1): $m \mid (a - b) + (c - d) = (a + c) - (b + d)$, hence $a + c \equiv b + d \pmod{m}$.

For (2): Note $m \mid (a - b)$ implies $m \mid c(a - b)$, and $m \mid (c - d)$ implies $m \mid b(c - d)$. Add these divisibilities to get $m \mid c(a - b) + b(c - d) = ac - bd$, so $ac \equiv bd \pmod{m}$. □

0.6 L6: 2.1 Congruences, 2.2 Linear congruences (1 var.)

L6: Congruences

§ Remark (Proposition Proposition 2.4 (Arithmetic with congruences) in class terms) Recall that Proposition 2.4 (Arithmetic with congruences) from the previous lecture can be expressed in terms of equivalence classes modulo m: for residue classes $[a], [c] \in \mathbb{Z}_m$,

$$[a] + [c] = [a + c] \qquad \text{and} \qquad [a][c] = [ac].$$

This viewpoint allows construction of addition and multiplication tables for residue classes (modulo m) using any complete residue system.

Examples: tables for $\mathbb{Z}_4$ and $\mathbb{Z}_5$

$+$	$[0]_4$	$[1]_4$	$[2]_4$	$[3]_4$
$[0]_4$	$[0]_4$	$[1]_4$	$[2]_4$	$[3]_4$
$[1]_4$	$[1]_4$	$[2]_4$	$[3]_4$	$[0]_4$
$[2]_4$	$[2]_4$	$[3]_4$	$[0]_4$	$[1]_4$
$[3]_4$	$[3]_4$	$[0]_4$	$[1]_4$	$[2]_4$

$\cdot$	$[0]_4$	$[1]_4$	$[2]_4$	$[3]_4$
$[0]_4$	$[0]_4$	$[0]_4$	$[0]_4$	$[0]_4$
$[1]_4$	$[0]_4$	$[1]_4$	$[2]_4$	$[3]_4$
$[2]_4$	$[0]_4$	$[2]_4$	$[0]_4$	$[2]_4$
$[3]_4$	$[0]_4$	$[3]_4$	$[2]_4$	$[1]_4$

§ Remark (structure of $\mathbb{Z}_4$) Set $\mathbb{Z}_4 := \{[0]_4, [1]_4, [2]_4, [3]_4\}$. With the above addition and multiplication it is a ring, but not a field: not every nonzero element has a multiplicative inverse (e.g., $[2]_4$ is a zero divisor).

$+$	$[0]_5$	$[1]_5$	$[2]_5$	$[3]_5$	$[4]_5$
$[0]_5$	$[0]_5$	$[1]_5$	$[2]_5$	$[3]_5$	$[4]_5$
$[1]_5$	$[1]_5$	$[2]_5$	$[3]_5$	$[4]_5$	$[0]_5$
$[2]_5$	$[2]_5$	$[3]_5$	$[4]_5$	$[0]_5$	$[1]_5$
$[3]_5$	$[3]_5$	$[4]_5$	$[0]_5$	$[1]_5$	$[2]_5$
$[4]_5$	$[4]_5$	$[0]_5$	$[1]_5$	$[2]_5$	$[3]_5$

$\cdot$	$[0]_5$	$[1]_5$	$[2]_5$	$[3]_5$	$[4]_5$
$[0]_5$	$[0]_5$	$[0]_5$	$[0]_5$	$[0]_5$	$[0]_5$
$[1]_5$	$[0]_5$	$[1]_5$	$[2]_5$	$[3]_5$	$[4]_5$
$[2]_5$	$[0]_5$	$[2]_5$	$[4]_5$	$[1]_5$	$[3]_5$
$[3]_5$	$[0]_5$	$[3]_5$	$[1]_5$	$[4]_5$	$[2]_5$
$[4]_5$	$[0]_5$	$[4]_5$	$[3]_5$	$[2]_5$	$[1]_5$

§ Remark (structure of $\mathbb{Z}_5$) Let $\mathbb{Z}_5 := \{[0]_5, [1]_5, [2]_5, [3]_5, [4]_5\}$. Because 5 is prime every nonzero class has a multiplicative inverse, so $\mathbb{Z}_5$ is a field.

§ Question (why extra structure?) Why does $\mathbb{Z}_5$ have more structure (a field) than $\mathbb{Z}_4$ (only a ring)?

§ Answer (primality of modulus) Essentially: $\mathbb{Z}_m$ is a field if and only if m is prime. If m is composite, then some nonzero classes are zero divisors (no inverses). If $m = p$ is prime, then every nonzero class has an inverse (standard proof via Bezout's identity).

Cancellation and a useful proposition

§ Proposition 2.5 (Cancellation in congruences) Let $a, b, c, m \in \mathbb{Z}$ with $m > 0$, and let $d = \gcd(c, m)$. Then

$$ca \equiv cb \pmod{m} \iff a \equiv b \pmod{m/d}.$$

Proof. Suppose $ca \equiv cb \pmod{m}$, i.e. $m \mid c(a-b)$. Let $d = \gcd(c, m)$. Write $c = d \cdot c'$ and $m = d \cdot m'$ so that $\gcd(c', m') = 1$. From $m \mid c(a-b)$ we get $dm' \mid dc'(a-b)$, hence $m' \mid c'(a-b)$. Since $\gcd(c', m') = 1$, it follows that $m' \mid (a-b)$, i.e. $a \equiv b \pmod{m'}$, which is $a \equiv b \pmod{m/d}$.

Conversely, if $a \equiv b \pmod{m/d}$ then $m/d \mid (a-b)$, so $m/d \cdot d = m \mid d(a-b)$. But $d \mid c$, so $c(a-b)$ is a multiple of $d(a-b)$, hence $m \mid c(a-b)$ and $ca \equiv cb \pmod{m}$. This proves the equivalence. ☐

Linear congruences in one variable

§ Definition (linear congruence) A congruence of the form

$$ax \equiv b \pmod{m},$$

with unknown $x \in \mathbb{Z}$ and given integers a, b, m with $m > 0$, is called a *linear congruence* in the variable x.

§ Examples (small checks)

- $2x \equiv 3 \pmod 4$ has no solution: checking $x \equiv 0, 1, 2, 3 \pmod 4$ yields no solution.
- $2x \equiv 3 \pmod 5$ has the unique solution $x \equiv 4 \pmod 5$.
- $2x \equiv 4 \pmod 6$ has two incongruent solutions modulo 6: $x \equiv 2, 5 \pmod 6$.
- $3x \equiv 9 \pmod 6$ has three incongruent solutions modulo 6: $x \equiv 1, 3, 5 \pmod 6$.

§ Theorem 2.6 (Solvability and number of solutions of linear congruences) Let $a, b, m \in \mathbb{Z}$ with $m > 0$ and let $d = \gcd(a, m)$. Then:

1. The congruence $ax \equiv b \pmod m$ has a solution iff $d \mid b$.
2. If $d \mid b$ then there are exactly d incongruent solutions modulo m. Concretely, if x_0 is one solution, then all solutions are
$$x = x_0 + \frac{m}{d}n, \qquad n \in \mathbb{Z},$$
and the classes $x_0 + \frac{m}{d}n$ for $n = 0, 1, \ldots, d-1$ give a complete set of incongruent solutions modulo m.

Proof. We use the standard equivalence between the congruence and a linear Diophantine equation. The congruence $ax \equiv b \pmod m$ is equivalent to the existence of an integer y such that

$$ax - my = b. \tag{*}$$

Thus solvability of the congruence is equivalent to solvability of the linear Diophantine equation $(*)$.

(1) If $(*)$ has an integer solution, then $d = \gcd(a, m)$ divides the left-hand side $ax - my$ and hence divides b. Conversely, if $d \mid b$, write $b = de$ for some $e \in \mathbb{Z}$. By Bezout's identity (see **??** or the standard result), there exist integers r, s with $ar + ms = d$. Multiply both sides by e to obtain $are + mse = de = b$. Then $x_0 = re$ and $y_0 = -se$ solve $(*)$, so a solution exists.

(2) Fix any solution x_0 of $ax \equiv b \pmod m$. For any integer n consider

$$x = x_0 + \frac{m}{d}n.$$

Then
$$ax = ax_0 + a\frac{m}{d}n = ax_0 + \frac{a}{d}mn \equiv ax_0 \equiv b \pmod m,$$
so each such x is a solution. Thus there are infinitely many integer solutions, all congruent modulo $\frac{m}{d}$ shifts.

Now suppose x is any solution. Then subtracting the equation for x_0 from the equation for x yields

$$a(x - x_0) \equiv 0 \pmod m,$$

i.e. $m \mid a(x - x_0)$. Write $a = da'$ and $m = dm'$, with $\gcd(a', m') = 1$. Then $m'd \mid da'(x - x_0)$, so $m' \mid a'(x - x_0)$. Since $\gcd(a', m') = 1$ we deduce $m' \mid (x - x_0)$, hence $x - x_0 = m'n = \frac{m}{d}n$ for some $n \in \mathbb{Z}$. This shows every solution has the stated form. Two such solutions
$$x_0 + \frac{m}{d}n_1 \quad \text{and} \quad x_0 + \frac{m}{d}n_2$$
are congruent modulo m iff
$$\frac{m}{d}(n_1 - n_2) \equiv 0 \pmod m \iff m \mid \frac{m}{d}(n_1 - n_2).$$

Dividing both sides by m/d yields $d \mid (n_1 - n_2)$, so $n_1 \equiv n_2 \pmod{d}$. Therefore choosing $n = 0, 1, \ldots, d-1$ produces exactly d incongruent solutions modulo m. ☐

§ Corollary 2.7 (Explicit solution set) Under the hypotheses of Theorem Theorem 2.6 (Solvability and number of solutions of linear congruences), if $d = \gcd(a, m)$ and $d \mid b$, then the d incongruent solutions modulo m are

$$x_0 + \frac{m}{d}n, \qquad n = 0, 1, \ldots, d-1,$$

where x_0 is any particular solution.

Worked example

§ Example (solve $16x \equiv 8 \pmod{28}$) We solve $16x \equiv 8 \pmod{28}$. Compute $d = \gcd(16, 28)$.

Euclidean algorithm:

$$28 = 1 \cdot 16 + 12, \qquad 16 = 1 \cdot 12 + 4, \qquad 12 = 3 \cdot 4 + 0,$$

so $d = 4$. Since $4 \mid 8$, the congruence has $d = 4$ incongruent solutions modulo 28 by Theorem Theorem 2.6 (Solvability and number of solutions of linear congruences).

One method: divide the congruence by $d = 4$ to get (working modulo $28/4 = 7$)

$$4x \equiv 2 \pmod{7}.$$

Since $4^{-1} \equiv 2 \pmod{7}$ (because $4 \cdot 2 = 8 \equiv 1 \pmod{7}$), multiply both sides by 2:

$$x \equiv 4 \pmod{7}.$$

Thus the solutions modulo 28 are

$$x \equiv 4 + 7n, \qquad n = 0, 1, 2, 3,$$

which gives the incongruent solutions $x \equiv 4, 11, 18, 25 \pmod{28}$.

(Alternatively, one can find an explicit x_0 via the extended Euclidean computation: from $4 = 2 \cdot 16 - 1 \cdot 28$ we get $8 = 4 \cdot 16 - 2 \cdot 28$, so $x_0 = 4$ is a particular solution, yielding the same solution set.)

Multiplicative inverses

§ Definition (multiplicative inverse modulo m) If $a \in \mathbb{Z}$ and there exists $x \in \mathbb{Z}$ with $ax \equiv 1 \pmod{m}$, then x is called a *multiplicative inverse* of a modulo m (often denoted $a^{-1} \pmod{m}$).

§ Corollary 2.8 (When inverses exist) The congruence $ax \equiv 1 \pmod{m}$ has a solution iff $\gcd(a, m) = 1$. In that case the inverse is unique modulo m.

Proof. Apply Theorem Theorem 2.6 (Solvability and number of solutions of linear congruences) with $b = 1$. If $d = \gcd(a, m) > 1$ then $d \nmid 1$ so there is no solution. If $d = 1$ then there is exactly one solution modulo m. ☐

§ Example (solve $5x \equiv 12 \pmod{16}$) Since $\gcd(5, 16) = 1$, the congruence has a unique solution modulo 16 by Corollary Corollary 2.8 (When inverses exist). One checks that $5^{-1} \pmod{16} = 13$ because $5 \cdot 13 = 65 \equiv 1 \pmod{16}$. Multiplying by 13 gives

$$x \equiv 13 \cdot 12 \pmod{16}.$$

Compute $13 \cdot 12 = 156 \equiv 12 \pmod{16}$ (since $156 - 144 = 12$), so $x \equiv 12 \pmod{16}$.

0.7 L7: 2.3 Chinese Remainder Thm, 2.4 Wilson's Thm, 2.5 Fermat's Little Thm

L7: The Chinese Remainder Theorem, Wilson's Theorem, Fermat's Little Theorem

Example (motivation). Find a positive integer x with

$$x \equiv 2 \pmod 3, \qquad x \equiv 1 \pmod 4, \qquad x \equiv 3 \pmod 5.$$

This is a system of linear congruences in one variable.

§ Theorem 2.9 (Chinese Remainder Theorem) Let $m_1, m_2, \ldots, m_n$ be pairwise relatively prime positive integers, and let $b_1, \ldots, b_n$ be arbitrary integers. Then the system

$$x \equiv b_i \pmod{m_i} \qquad (i = 1, \ldots, n)$$

has a solution, and any two solutions are congruent modulo $M := m_1 m_2 \cdots m_n$. Hence the solution is unique modulo M.

Proof. Let $M := \prod_{i=1}^n m_i$ and define $M_i := M/m_i$. Since $(M_i, m_i) = 1$ by the pairwise coprimality hypothesis, by Corollary 2.8 (When inverses exist) (existence of inverses when gcd is 1) there exists x_i with

$$M_i x_i \equiv 1 \pmod{m_i}.$$

Put

$$x := \sum_{i=1}^n b_i M_i x_i.$$

For a fixed index i, reduce modulo m_i: every term $b_j M_j x_j$ with $j \neq i$ is a multiple of m_i (since $m_i \mid M_j$ for $j \neq i$), while $M_i x_i \equiv 1 \pmod{m_i}$ by construction. Thus $x \equiv b_i \pmod{m_i}$ for each i, so x is a solution.

If x' is another solution, then $m_i \mid (x - x')$ for every i, and since the m_i are pairwise coprime it follows that $M \mid (x - x')$. Hence $x \equiv x' \pmod M$, proving uniqueness modulo M. □

Example (continued). Here $m_1 = 3, m_2 = 4, m_3 = 5$, so $M = 60$ and

$$M_1 = 60/3 = 20, \qquad M_2 = 60/4 = 15, \qquad M_3 = 60/5 = 12.$$

Solve $M_i x_i \equiv 1 \pmod{m_i}$:

$$20x_1 \equiv 1 \pmod 3 \implies 2x_1 \equiv 1 \pmod 3 \implies x_1 \equiv 2 \pmod 3,$$

so choose $x_1 = 2$.

$$15x_2 \equiv 1 \pmod 4 \implies 3x_2 \equiv 1 \pmod 4 \implies x_2 \equiv 3 \pmod 4,$$

so choose $x_2 = 3$.

$$12x_3 \equiv 1 \pmod 5 \implies 2x_3 \equiv 1 \pmod 5 \implies x_3 \equiv 3 \pmod 5,$$

so choose $x_3 = 3$.

Now

$$x = b_1 M_1 x_1 + b_2 M_2 x_2 + b_3 M_3 x_3 = 2 \cdot 20 \cdot 2 + 1 \cdot 15 \cdot 3 + 3 \cdot 12 \cdot 3 = 80 + 45 + 108 = 233.$$

Hence $x \equiv 233 \equiv 53 \pmod{60}$, and the least positive solution is 53.

§ Lemma 2.10 (Self-inverse modulo a prime) Let p be a prime and $a \in \mathbb{Z}$. Then a is its own inverse modulo p (i.e. $a^2 \equiv 1 \pmod{p}$) if and only if $a \equiv \pm 1 \pmod{p}$.

Proof. If $a^2 \equiv 1 \pmod{p}$ then $p \mid (a-1)(a+1)$. Since p is prime, $p \mid (a-1)$ or $p \mid (a+1)$, so $a \equiv 1$ or $a \equiv -1 \pmod{p}$. The converse is immediate. □

§ Theorem 2.11 (Wilson's Theorem) Let p be a prime. Then

$$(p-1)! \equiv -1 \pmod{p}.$$

Proof. The result is trivial for $p = 2, 3$, so assume $p \geq 5$. The integers $1, 2, \ldots, p-1$ can be paired into multiplicative inverses modulo p. By Lemma 0.7, the only elements that are their own inverses are 1 and $p-1$. Thus, aside from 1 and $p-1$, the remaining $p-3$ elements pair off, and each pair multiplies to 1 modulo p. Therefore

$$(p-1)! \equiv 1 \cdot (\text{product of inverse-pairs}) \cdot (p-1) \equiv 1 \cdot 1 \cdots 1 \cdot (-1) \equiv -1 \pmod{p},$$

as required. □

§ Remark (converse) The converse is also true: if $n > 1$ and $(n-1)! \equiv -1 \pmod{n}$ then n must be prime.

§ Proposition 2.12 (Converse to Wilson's Theorem) Let $n > 1$ be an integer. If $(n-1)! \equiv -1 \pmod{n}$ then n is prime.

Proof. Suppose n is composite, so $n = ab$ with $1 < a \leq b < n$. Then $a \mid (n-1)!$ because $(n-1)!$ contains the factor a, and hence $(n-1)! \equiv 0 \pmod{a}$. The hypothesis $(n-1)! \equiv -1 \pmod{n}$ implies $(n-1)! \equiv -1 \pmod{a}$ (since $a \mid n$), so $-1 \equiv 0 \pmod{a}$, which is impossible. Therefore no such nontrivial factor a exists and n must be prime. □

§ Remark (Wilson primes) A prime p is called a *Wilson prime* if $(p-1)! \equiv -1 \pmod{p^2}$. Only three Wilson primes are known: $5, 13, 563$. It is unknown whether infinitely many (or only finitely many) Wilson primes exist.

§ Theorem 2.13 (Fermat's Little Theorem) Let p be a prime and let $a \in \mathbb{Z}$ with $p \nmid a$. Then

$$a^{p-1} \equiv 1 \pmod{p}.$$

Proof. Consider the $p-1$ integers $a, 2a, \ldots, (p-1)a$. None of these is congruent to $0 \pmod{p}$ (since $p \nmid a$), and no two are congruent modulo p (if $ia \equiv ja \pmod{p}$ with $1 \leq i < j \leq p-1$, multiply by the inverse of a modulo p, which exists by Corollary 2.8 (When inverses exist), to deduce $i \equiv j \pmod{p}$, impossible). Hence the residues of $a, 2a, \ldots, (p-1)a$ modulo p are a permutation of $1, 2, \ldots, p-1$. Multiplying all these congruences yields

$$a^{p-1}(p-1)! \equiv (p-1)! \pmod{p}.$$

Since $p \nmid (p-1)!$ (true for prime p), we may cancel $(p-1)!$ from both sides to obtain $a^{p-1} \equiv 1 \pmod{p}$. (Note: this cancellation step uses only that $\gcd((p-1)!, p) = 1$; Wilson's Theorem is not required.) □

0.8 L8: 2.5 Pseudoprimes, 2.6 Euler's Thm, 3.1 Arithmetic functions/multiplicativity

L8: Corollaries to Fermat's Little Theorem, Pseudoprimes, Euler's Theorem, Arithmetic Functions

§ Corollary 2.14 (Inverse modulo a prime) Let p be a prime and let $a \in \mathbb{Z}$ with $p \nmid a$. Then a^{p-2} is the multiplicative inverse of a modulo p; that is,

$$a \cdot a^{p-2} \equiv 1 \pmod{p}.$$

Proof. By Fermat's Little Theorem (Theorem 0.7), $a^{p-1} \equiv 1 \pmod{p}$. Multiply both sides by a^{-1} if you like, or simply rewrite $a^{p-1} = a \cdot a^{p-2}$ to get $a \cdot a^{p-2} \equiv 1 \pmod{p}$. Thus a^{p-2} is an inverse of a modulo p. □

§ Remark (application of Corollary Corollary 2.14 (Inverse modulo a prime)) Corollary Corollary 2.14 (Inverse modulo a prime) is useful for solving linear congruences $ax \equiv b \pmod{p}$ when p is prime and $p \nmid a$: multiply both sides by a^{p-2} to obtain the solution.

§ Corollary 2.15 (Fresh form of Fermat) Let p be prime and $a \in \mathbb{Z}$. Then

$$a^p \equiv a \pmod{p}.$$

Proof. If $p \mid a$ the congruence is trivial. If $p \nmid a$, multiply Fermat's Little Theorem $a^{p-1} \equiv 1 \pmod{p}$ by a to get $a^p \equiv a \pmod{p}$. □

§ Corollary 2.16 (case $a = 2$) For every prime p,

$$2^p \equiv 2 \pmod{p}.$$

Proof. Take $a = 2$ in Corollary Corollary 2.15 (Fresh form of Fermat). □

§ Question (converse for base 2) Is the converse of Corollary Corollary 2.16 (case $a = 2$) true? That is, if $n > 1$ and $2^n \equiv 2 \pmod{n}$, must n be prime?

§ Answer and counterexample (341) No. Example: $n = 341 = 11 \cdot 31$ is composite but satisfies $2^{341} \equiv 2 \pmod{341}$. Since $(11, 31) = 1$, it suffices to check modulo 11 and 31.

By Fermat's Little Theorem,

$$2^{10} \equiv 1 \pmod{11} \quad \Rightarrow \quad 2^{341} \equiv (2^{10})^{34} \cdot 2 \equiv 1^{34} \cdot 2 \equiv 2 \pmod{11},$$

and

$$2^{30} \equiv 1 \pmod{31} \quad \Rightarrow \quad 2^{341} \equiv (2^{30})^{11} \cdot 2^{11} \equiv 1^{11} \cdot 2^{11} \pmod{31}.$$

Since $2^5 \equiv -1 \pmod{31}$ (or compute directly), we have $2^{11} \equiv 2 \pmod{31}$, so $2^{341} \equiv 2 \pmod{31}$. Hence $2^{341} \equiv 2 \pmod{341}$ although 341 is composite.

§ Definition (pseudoprime to base 2) A composite integer $n > 1$ is called a *(base-2) pseudoprime* if

$$2^n \equiv 2 \pmod{n}.$$

§ Remark (pseudoprimes) 341 is the smallest base-2 pseudoprime. There are infinitely many pseudoprimes and they arise for different bases; pseudoprimes show that the naive converse of Fermat's Little Theorem is false.

§ Definition (Euler's phi-function) For $n \in \mathbb{Z}_{>0}$ the Euler totient function $\phi(n)$ is the number of integers $1 \le x \le n$ with $\gcd(x, n) = 1$; equivalently,

$$\phi(n) := \#\{x \in \{1, 2, \ldots, n\} : \gcd(x, n) = 1\}.$$

§ Examples (values of ϕ)

$$\phi(12) = 4 \quad (\text{the classes } 1, 5, 7, 11), \qquad \phi(14) = 6 \quad (\text{the classes } 1, 3, 5, 9, 11, 13),$$

and for a prime p, $\phi(p) = p - 1$ (classes $1, 2, \ldots, p-1$).

§ Theorem 2.17 (Euler's Theorem) Let $m > 0$ and $a \in \mathbb{Z}$ with $\gcd(a, m) = 1$. Then

$$a^{\phi(m)} \equiv 1 \pmod{m}.$$

Proof. Let $r_1, r_2, \ldots, r_{\phi(m)}$ be the positive integers $\le m$ that are relatively prime to m. Since $\gcd(a, m) = 1$, multiplication by a permutes this reduced residue system: the numbers $ar_1, \ldots, ar_{\phi(m)}$ are all distinct modulo m and each is relatively prime to m (see Lemma 1.5 and Lemma 1.14 referenced earlier). Hence the set $\{ar_i \bmod m\}$ is the same set of residues as $\{r_i\}$. Multiplying all residues yields

$$a^{\phi(m)} \prod_{i=1}^{\phi(m)} r_i \equiv \prod_{i=1}^{\phi(m)} r_i \pmod{m}.$$

Since $\gcd\left(\prod_i r_i,\, m\right) = 1$, we may cancel the product to obtain $a^{\phi(m)} \equiv 1 \pmod{m}$. □

§ Remark (special case) Fermat's Little Theorem is the special case $m = p$ prime, since $\phi(p) = p - 1$.

§ Definition (reduced residue system) A set of $\phi(m)$ integers each relatively prime to m and pairwise incongruent modulo m is called a *reduced residue system* modulo m.

§ Example (reduced residue systems) $\{1, 5, 7, 11\}$ is a reduced residue system modulo 12. If a is coprime to m, then $\{ar_1, \ldots, ar_{\phi(m)}\}$ is also a reduced residue system modulo m (this was the idea used in the proof of Euler's Theorem).

§ Corollary 2.19 (inverse via Euler) Let $m > 0$ and $a \in \mathbb{Z}$ with $\gcd(a, m) = 1$. Then $a^{\phi(m)-1}$ is the multiplicative inverse of a modulo m.

Proof. From $a^{\phi(m)} \equiv 1 \pmod{m}$ multiply by a^{-1} (or rewrite as $a \cdot a^{\phi(m)-1} \equiv 1$) to see $a^{\phi(m)-1}$ is an inverse of a modulo m. □

§ Remark (use of Corollary Corollary 2.19 (inverse via Euler)) Corollary Corollary 2.19 (inverse via Euler) helps solve linear congruences $ax \equiv b \pmod{m}$ when $\gcd(a, m) = 1$: multiply both sides by $a^{\phi(m)-1}$ to obtain the solution.

§ Definition (arithmetic function) An *arithmetic function* is a function $f : \mathbb{Z}_{>0} \to \mathbb{C}$ (or $\mathbb{R}$, or $\mathbb{Z}$). Examples: the Euler ϕ-function, the divisor-counting function $d(n)$, the sum-of-divisors function $\sigma(n)$, etc.

§ Definition (multiplicative vs completely multiplicative) An arithmetic function f is *multiplicative* if $f(mn) = f(m)f(n)$ whenever $\gcd(m, n) = 1$. It is *completely multiplicative* if $f(mn) = f(m)f(n)$ for all positive integers m, n (no coprimality required).

§ Proposition (values determined by prime powers) If f is multiplicative and

$$n = p_1^{a_1} p_2^{a_2} \cdots p_r^{a_r}$$

is the prime factorization of n, then

$$f(n) = \prod_{i=1}^{r} f\big(p_i^{a_i}\big).$$

If f is completely multiplicative, then

$$f(n) = \prod_{i=1}^{r} f(p_i)^{a_i}.$$

Proof. Write n as a product of relatively prime prime-power factors and apply multiplicativity repeatedly. □

§ Remark (implication) Thus knowing a multiplicative arithmetic function on prime powers (or a completely multiplicative function on primes) determines it everywhere.

0.9 L9: 3.1 Arithmetic functions/multiplicativity, 3.2 Euler phi function

L9: Arithmetic Functions and Multiplicativity

§ Summation over divisors: notation By

$$\sum_{d|n} f(d)$$

we mean the sum of $f(d)$ over all *positive* divisors d of n. For example,

$$\sum_{d|12} f(d) = f(1) + f(2) + f(3) + f(4) + f(6) + f(12).$$

§ Multiplicative functions and divisor sums Let f be an arithmetic function and define, for $n \in \mathbb{Z}_{>0}$,

$$F(n) := \sum_{d|n} f(d).$$

If f is multiplicative then F is multiplicative.

Proof. Let $m, n \in \mathbb{Z}_{>0}$ with $\gcd(m, n) = 1$. We must show $F(mn) = F(m)\,F(n)$.

Claim: Every positive divisor d of mn can be written uniquely as $d = d_1 d_2$ with $d_1 \mid m$, $d_2 \mid n$ and $\gcd(d_1, d_2) = 1$.

Proof of claim: Write the prime factorizations

$$m = \prod_{i=1}^{r} p_i^{a_i}, \qquad n = \prod_{j=1}^{s} q_j^{b_j},$$

where the primes p_i and q_j are distinct because $\gcd(m, n) = 1$. Then any divisor d of mn has the form

$$d = \prod_{i=1}^{r} p_i^{e_i} \prod_{j=1}^{s} q_j^{f_j},$$

with $0 \le e_i \le a_i$ and $0 \le f_j \le b_j$. Set

$$d_1 = \prod_{i=1}^{r} p_i^{e_i}, \qquad d_2 = \prod_{j=1}^{s} q_j^{f_j}.$$

Clearly $d_1 \mid m$, $d_2 \mid n$, $\gcd(d_1, d_2) = 1$, and $d = d_1 d_2$. Uniqueness follows from uniqueness of prime factorization. □

Using the claim,

$$F(mn) = \sum_{d|mn} f(d) = \sum_{\substack{d_1|m \\ d_2|n}} f(d_1 d_2).$$

Since f is multiplicative and $\gcd(d_1, d_2) = 1$ for every pair occurring in the sum, we have $f(d_1 d_2) = f(d_1) f(d_2)$. Thus

$$F(mn) = \sum_{d_1|m} \sum_{d_2|n} f(d_1) f(d_2) = \Big(\sum_{d_1|m} f(d_1)\Big)\Big(\sum_{d_2|n} f(d_2)\Big) = F(m)F(n).$$

This proves that F is multiplicative. □

§ Example: verification for $m = 3$, $n = 4$ Let $m = 3$ and $n = 4$. The positive divisors are

$$\{\, d_1 \mid 3 \,\} = \{1, 3\}, \qquad \{\, d_2 \mid 4 \,\} = \{1, 2, 4\}.$$

All divisors of 12 are the products $d_1 d_2$ with $d_1 \in \{1,3\}$ and $d_2 \in \{1,2,4\}$; hence

$$\sum_{d|12} f(d) = \sum_{d_1|3} \sum_{d_2|4} f(d_1 d_2) = \sum_{d_1|3} \sum_{d_2|4} f(d_1) f(d_2)$$

(using multiplicativity of f), so

$$\sum_{d|12} f(d) = \Big(\sum_{d_1|3} f(d_1)\Big)\Big(\sum_{d_2|4} f(d_2)\Big) = F(3)F(4).$$

§ The Euler φ-function is multiplicative The Euler totient function $\varphi(n)$ (the number of integers in $\{1, 2, \ldots, n\}$ relatively prime to n) is multiplicative: if $\gcd(m, n) = 1$ then

$$\varphi(mn) = \varphi(m)\,\varphi(n).$$

Proof. List the integers $1, 2, \ldots, mn$ in an $m \times n$ array with rows indexed by $i = 1, \ldots, m$ and columns by $k = 0, \ldots, n-1$ as

$$\begin{array}{ccccc}
1 & m+1 & 2m+1 & \cdots & (n-1)m+1 \\
2 & m+2 & 2m+2 & \cdots & (n-1)m+2 \\
\vdots & \vdots & \vdots & & \vdots \\
m & m+m & 2m+m & \cdots & (n-1)m+m
\end{array}$$

Consider the i-th row

$$i,\ m+i,\ 2m+i,\ \ldots,\ (n-1)m+i.$$

If $\gcd(i, m) > 1$ then every entry of that row shares a common factor with m, and hence cannot be relatively prime to mn. Thus only rows with $\gcd(i, m) = 1$ can contain integers relatively prime to mn. There are precisely $\varphi(m)$ such rows.

Fix a row with $\gcd(i, m) = 1$. The n entries of that row are distinct modulo n: if $jm+i \equiv km+i \pmod{n}$ with $0 \le j, k \le n-1$, then $(j-k)m \equiv 0 \pmod{n}$. Since $\gcd(m, n) = 1$, m has a multiplicative inverse modulo n, so $j \equiv k \pmod{n}$, hence $j = k$. Thus the row is a complete residue system modulo n, so exactly $\varphi(n)$ of its entries are relatively prime to n. Because those entries are relatively prime to both m and n, they are relatively prime to mn.

Therefore each of the $\varphi(m)$ admissible rows contains exactly $\varphi(n)$ integers relatively prime to mn. Thus

$$\varphi(mn) = \varphi(m)\varphi(n).$$

□

§ Values of φ at prime powers Let p be prime and $a \in \mathbb{Z}_{>0}$. Then

$$\varphi(p^a) = p^a - p^{a-1} = p^a\Big(1 - \frac{1}{p}\Big).$$

Proof. Among the p^a integers $1, 2, \ldots, p^a$, those not coprime to p^a are exactly the multiples of p:

$$p, 2p, 3p, \ldots, p^{a-1}p,$$

and there are p^{a-1} such multiples. Hence

$$\varphi(p^a) = p^a - p^{a-1} = p^a\Big(1 - \frac{1}{p}\Big).$$

□

§ Product formula for $\varphi(n)$ For $n \in \mathbb{Z}_{>0}$,

$$\varphi(n) = n \prod_{p \mid n} \Big(1 - \frac{1}{p}\Big),$$

where the product is taken over the distinct prime divisors p of n.

Proof. If $n = 1$ the result holds (empty product $= 1$). For $n > 1$, write the prime factorization

$$n = \prod_{i=1}^{r} p_i^{a_i}.$$

By Theorems The Euler φ-function is multiplicative and Values of φ at prime powers,

$$\varphi(n) = \prod_{i=1}^{r} \varphi(p_i^{a_i}) = \prod_{i=1}^{r} p_i^{a_i}\Big(1 - \frac{1}{p_i}\Big) = n \prod_{p \mid n} \Big(1 - \frac{1}{p}\Big).$$

□

§ Example: $\varphi(504)$ Since $504 = 2^3 \cdot 3^2 \cdot 7$, we get

$$\varphi(504) = 504\Big(1 - \frac{1}{2}\Big)\Big(1 - \frac{1}{3}\Big)\Big(1 - \frac{1}{7}\Big).$$

Compute:

$$\Big(1 - \frac{1}{2}\Big)\Big(1 - \frac{1}{3}\Big)\Big(1 - \frac{1}{7}\Big) = \frac{1}{2} \cdot \frac{2}{3} \cdot \frac{6}{7} = \frac{2}{7},$$

so $\varphi(504) = 504 \cdot \frac{2}{7} = 144$.

§ Gauss's identity for φ For $n \in \mathbb{Z}_{>0}$,

$$\sum_{d \mid n} \varphi(d) = n.$$

Proof. For each positive divisor d of n define

$$S_d := \{\, m \in \mathbb{Z} : 1 \le m \le n,\ \gcd(m, n) = d \,\}.$$

Note that $\gcd(m, n) = d$ if and only if $\gcd(m/d, n/d) = 1$, so the number of integers in S_d equals $\varphi(n/d)$. The sets S_d (as d runs over the positive divisors of n) partition the set $\{1, 2, \ldots, n\}$; hence

$$n = \sum_{d \mid n} |S_d| = \sum_{d \mid n} \varphi(n/d).$$

Reindexing the sum by replacing d with n/d shows

$$n = \sum_{d \mid n} \varphi(d),$$

as claimed. □

§ Example: $n = 12$ For $n = 12$ the divisors and corresponding sets are

$$
\begin{aligned}
S_1 &= \{1,5,7,11\}, & |S_1| &= \varphi(12) = 4,\\
S_2 &= \{2,10\}, & |S_2| &= \varphi(6) = 2,\\
S_3 &= \{3,9\}, & |S_3| &= \varphi(4) = 2,\\
S_4 &= \{4,8\}, & |S_4| &= \varphi(3) = 2,\\
S_6 &= \{6\}, & |S_6| &= \varphi(2) = 1,\\
S_{12} &= \{12\}, & |S_{12}| &= \varphi(1) = 1.
\end{aligned}
$$

Summing gives

$$\sum_{d|12} \varphi(d) = 4+2+2+2+1+1 = 12,$$

confirming Gauss's identity.

0.10 L10: 3.3 Divisor function, 3.4 Sum of divisors, 3.5 Perfect numbers, 3.6 Möbius function

L10: The number of positive divisors function

§ The number of positive divisors function Let $n \in \mathbb{Z}_{>0}$. The number of positive divisors of n (the divisor-count function) is the function

$$V(n) := \#\{\, d \in \mathbb{Z}_{>0} : d \mid n \,\}.$$

§ Multiplicativity of the divisor-count function The function $V(n)$ is multiplicative: if $\gcd(m, n) = 1$ then

$$V(mn) = V(m)\,V(n).$$

Proof. Write

$$V(n) = \sum_{d \mid n} 1,$$

the sum of the constant arithmetic function $f(d) \equiv 1$ over positive divisors of n. The constant function $f(d) = 1$ is multiplicative, so by Theorem Multiplicative functions and divisor sums (divisor-sum preserving multiplicativity) the divisor-sum $V(n)$ is multiplicative. □

§ Values of V at prime powers Let p be prime and $a \in \mathbb{Z}_{\geq 0}$. Then

$$V\big(p^a\big) = a + 1.$$

Proof. The positive divisors of p^a are exactly

$$1, p, p^2, \ldots, p^a,$$

so there are $a + 1$ of them. □

§ Prime-power formula for $V(n)$ If

$$n = \prod_{i=1}^{r} p_i^{a_i}$$

is the prime factorization of n (distinct primes p_i, exponents $a_i \geq 0$), then

$$V(n) = \prod_{i=1}^{r} (a_i + 1).$$

Proof. Since V is multiplicative (Theorem Multiplicativity of the divisor-count function), and each $V(p_i^{a_i}) = a_i + 1$ (Theorem Values of V at prime powers), multiplicativity yields

$$V(n) = \prod_{i=1}^{r} V(p_i^{a_i}) = \prod_{i=1}^{r} (a_i + 1).$$

□

§ Example: $V(504)$ If $504 = 2^3 \cdot 3^2 \cdot 7^1$ then

$$V(504) = (3 + 1)(2 + 1)(1 + 1) = 4 \cdot 3 \cdot 2 = 24.$$

§ The sum-of-divisors function σ For $n \in \mathbb{Z}_{>0}$ define

$$\sigma(n) := \sum_{d|n} d,$$

the sum of the positive divisors of n.

§ Multiplicativity of σ The function $\sigma(n)$ is multiplicative: if $\gcd(m, n) = 1$ then

$$\sigma(mn) = \sigma(m)\,\sigma(n).$$

Proof. This follows from Theorem Multiplicative functions and divisor sums by observing that the arithmetic function $f(d) = d$ is multiplicative: if $\gcd(a, b) = 1$ then $(ab) = a \cdot b$ and summing multiplicatively gives the result. (Equivalently, check $f(ab) = ab = f(a)f(b)$ for coprime a, b.) □

§ Values of σ at prime powers Let p be prime and $a \in \mathbb{Z}_{\geq 0}$. Then

$$\sigma\big(p^a\big) = 1 + p + \cdots + p^a = \frac{p^{a+1} - 1}{p - 1}.$$

Proof. The divisors of p^a are $1, p, \ldots, p^a$, so the sum is the finite geometric series $1 + p + \cdots + p^a = (p^{a+1} - 1)/(p - 1)$. □

§ Prime-power product formula for $\sigma(n)$ If $n = \prod_{i=1}^{r} p_i^{a_i}$ then

$$\sigma(n) = \prod_{i=1}^{r} \frac{p_i^{a_i+1} - 1}{p_i - 1}.$$

Proof. By multiplicativity (Theorem Multiplicativity of σ) and Theorem Values of σ at prime powers applied to each prime power factor. □

§ Example: $\sigma(504)$ For $504 = 2^3 \cdot 3^2 \cdot 7$,

$$\sigma(504) = \Big(\frac{2^4 - 1}{2 - 1}\Big)\Big(\frac{3^3 - 1}{3 - 1}\Big)\Big(\frac{7^2 - 1}{7 - 1}\Big) = 15 \cdot 13 \cdot 8 = 1560.$$

§ Perfect numbers (definition) A positive integer n is *perfect* if $\sigma(n) = 2n$. Equivalently, the sum of the proper divisors of n (i.e., divisors $< n$) equals n.

§ Classical characterization of even perfect numbers An integer $n > 0$ is an even perfect number if and only if

$$n = 2^{p-1}\big(2^p - 1\big),$$

where p is prime and $2^p - 1$ is prime (a Mersenne prime).

Proof. We give both directions (Euler $\Rightarrow$ and Euclid $\Leftarrow$).

($\Leftarrow$) **(Euclid).** Suppose p and $2^p - 1$ are prime and set $n = 2^{p-1}(2^p - 1)$. Using multiplicativity of σ,

$$\sigma(n) = \sigma\big(2^{p-1}\big)\sigma(2^p - 1) = \frac{2^p - 1}{2 - 1} \cdot (1 + 2^p - 1) = (2^p - 1) \cdot 2^p = 2n,$$

since $\sigma(2^{p-1}) = 1+2+\cdots+2^{p-1} = (2^p-1)$ and $\sigma(2^p-1) = 1+(2^p-1) = 2^p$ (because 2^p-1 is prime). Hence n is perfect.

($\Rightarrow$) **(Euler).** Let n be an even perfect number. Write $n = 2^r m$ with $r \geq 1$ and m odd. Then multiplicativity yields

$$\sigma(n) = \sigma(2^r)\sigma(m) = (2^{r+1}-1)\sigma(m).$$

Since n is perfect, $\sigma(n) = 2n = 2^{r+1}m$. Thus

$$(2^{r+1}-1)\sigma(m) = 2^{r+1}m.$$

Because $\gcd(2^{r+1}-1, 2^{r+1}) = 1$, it follows that $2^{r+1} \mid \sigma(m)$. Hence write $\sigma(m) = 2^{r+1}c$ for some $c \in \mathbb{Z}_{>0}$, and substituting back gives

$$(2^{r+1}-1)2^{r+1}c = 2^{r+1}m \quad\Longrightarrow\quad m = (2^{r+1}-1)c.$$

We now show $c = 1$. If $c > 1$ then m has at least the divisors $1, c, m$ with $1 < c < m$, so

$$\sigma(m) \geq 1+c+m = 1+c+(2^{r+1}-1)c = 1+(2^{r+1})c,$$

whence $\sigma(m) \geq 1+2^{r+1}c$, contradicting $\sigma(m) = 2^{r+1}c$. Thus $c = 1$ and $m = 2^{r+1}-1$. Then $\sigma(m) = m+1 = 2^{r+1}$, so m is prime and $m = 2^{r+1}-1$ is a Mersenne prime. Setting $p = r+1$ gives $n = 2^{p-1}(2^p-1)$ with p prime and 2^p-1 prime, as required. $\square$

§ Remarks on perfect numbers - This theorem establishes a bijection between Mersenne primes and even perfect numbers. It is unknown whether there are infinitely many Mersenne primes (hence unknown whether there are infinitely many even perfect numbers). - It is also unknown whether any odd perfect numbers exist; the conjecture "every perfect number is even" remains open.

§ The Möbius function μ The Möbius function $\mu : \mathbb{Z}_{>0} \to \{-1, 0, 1\}$ is defined by

$$\mu(n) = \begin{cases} 1, & n = 1, \\ 0, & \text{if } p^2 \mid n \text{ for some prime } p, \\ (-1)^r, & \text{if } n = p_1p_2\cdots p_r \text{ is a product of } r \text{ distinct primes.} \end{cases}$$

§ Examples for μ Since $504 = 2^3 \cdot 3^2 \cdot 7$ contains a squared prime factor, $\mu(504) = 0$. Since $30 = 2\cdot 3\cdot 5$ is a product of three distinct primes, $\mu(30) = (-1)^3 = -1$.

§ Multiplicativity of μ The Möbius function $\mu(n)$ is multiplicative: if $\gcd(m,n) = 1$ then $\mu(mn) = \mu(m)\mu(n)$.

Proof. If either m or n is divisible by a square of a prime then so is mn, and both sides are 0. Otherwise write $m = \prod_{i=1}^r p_i$ and $n = \prod_{j=1}^s q_j$ as products of distinct primes (with the two lists disjoint by coprimality). Then

$$\mu(mn) = (-1)^{r+s} = (-1)^r(-1)^s = \mu(m)\mu(n).$$

$\square$

§ Möbius summatory identity (Proposition) For $n \in \mathbb{Z}_{>0}$,

$$\sum_{d\mid n} \mu(d) = \begin{cases} 1, & n = 1, \\ 0, & n > 1. \end{cases}$$

Proof. Define $F(n) = \sum_{d|n} \mu(d)$. The function F is multiplicative since it is a divisor-sum of a multiplicative function μ (Theorems Multiplicativity of μ and Multiplicative functions and divisor sums). Thus F is determined by its values on prime powers. Clearly $F(1) = 1$. If p is prime and $a \geq 1$ then

$$F(p^a) = \mu(1) + \mu(p) + \mu(p^2) + \cdots + \mu(p^a) = 1 + (-1) + 0 + \cdots + 0 = 0.$$

Hence $F(n) = 0$ for every $n > 1$ (by multiplicativity and prime-power vanishing), proving the claim. □

§ Example: $n = 12$ **(check)** For $n = 12$ the divisors are $1, 2, 3, 4, 6, 12$ and

$$\mu(1) + \mu(2) + \mu(3) + \mu(4) + \mu(6) + \mu(12) = 1 + (-1) + (-1) + 0 + 1 + 0 = 0,$$

consistent with Proposition Möbius summatory identity (Proposition).